UNIT 2 · TOPIC 2.4

Physical Detection Systems

Quiz, Answer Key

Teacher answer key, 18 questions, 1 point each. The correct answer is highlighted; the explanation and distractor notes follow.

1. Which of the following is a detective physical security control?

- ○ A. A deadbolt lock on a server room door
- ○ B. Full-disk encryption on all workstations
- **● C. A motion-activated CCTV camera inside the server room**
- ○ D. A multi-factor authentication system on network accounts

Correct answer: C

Why: A motion-activated CCTV camera detects and records movement, making it a detective control. Deadbolt locks are preventive (they block access). Full-disk encryption and MFA protect data and logical access respectively, neither monitors for physical intrusion events.

*Why the others are wrong: **A**, A deadbolt is preventive, it stops unauthorized entry rather than detecting it. Students often conflate physical barriers with detection. **B**, Encryption protects data confidentiality; it generates no alert or evidence of a physical intrusion. **D**, MFA is a logical access control and operates in the authentication layer, not the physical detection layer.*

2. A badge reader at the entrance of Xtensr Labs records every access attempt, including attempts where the card is rejected. What type of security control does this logging capability represent?

- ○ A. Preventive
- ○ B. Deterrent
- ○ C. Corrective
- **● D. Detective**

Correct answer: D

Why: Logging all access attempts, including failures, is a detective control: it creates an audit trail that can reveal suspicious patterns (e.g., repeated failed attempts before a successful entry). A deterrent discourages attacks, a preventive control blocks them, and a corrective control repairs damage after an event.

*Why the others are wrong: **A**, The badge reader also acts as a preventive control by denying entry, but the logging function specifically is detective. Students may conflate the physical door mechanism with the logging feature. **B**, Deterrents discourage attackers (e.g., a visible camera or warning sign), but a log file that is not visible to attackers does not deter. **C**, Corrective controls restore systems after an incident, not the function of an access log.*

3. Which of the following best describes the purpose of a tamper-evident seal placed on a server chassis?

- ○ A. It prevents unauthorized users from logging in to the server remotely.
- **● B. It provides a visible indication that the chassis may have been physically opened or altered.**
- ○ C. It encrypts the data stored on internal drives.
- ○ D. It automatically alerts the network operations center when removed.

Correct answer: B

Why: A tamper-evident seal is a detective control: when broken or disturbed, it visually signals that a physical access event may have occurred, supporting later investigation. It does not prevent access, encrypt data, or generate automated alerts by itself.

*Why the others are wrong: **A**, Remote login controls are logical, not physical detective controls, unrelated to a chassis seal. **C**, Encryption is a data confidentiality control; a tamper seal provides no cryptographic protection. **D**, A basic tamper-evident seal is passive (visual only). An active tamper sensor with alerting is a separate, more advanced technology, students may reasonably assume seals are "smart."*

4. While reviewing access logs for Xtensr Labs' new facility, a security analyst notices the same employee badge was used at the main entrance and at a rear server room 18 seconds apart, a distance that takes at least 4 minutes to walk. What detection technique is the analyst applying, and what does the finding suggest?

○ A. Perimeter monitoring; the badge reader hardware is malfunctioning and timestamps are unreliable.

○ B. Network traffic analysis; a replay attack is being launched against the access control system.

○ C. Biometric analysis; the employee's identity could not be verified at one of the readers.

● D. Access log anomaly detection; the badge may be cloned or shared, indicating a potential physical security breach.

Correct answer: D

Why: This is access log anomaly detection, identifying events that are physically or logically impossible under normal conditions. Two badge reads at distant locations within 18 seconds indicates the credential is likely cloned or being shared, both of which signal a potential physical security threat. The analyst is applying LO 2.4.C: using detection techniques to identify attacks.

*Why the others are wrong: **A**, A hardware malfunction is possible but is the less security-relevant interpretation; anomaly detection requires investigating the worst-case scenario first. **B**, A replay attack targets authentication packets on a network, not a physical badge event that was swiped at two locations. **C**, Biometric analysis involves physiological data; access log review is a separate technique. Students may conflate smart badge systems with biometrics.*

5. Xtensr Labs installs pressure sensors beneath the floor tiles of its server room. A security engineer explains that the sensors will trigger an alert if weight is detected in the room when no badge access has been logged. What type of physical attack is this detection method specifically designed to identify?

● A. Unauthorized physical intrusion through a path that bypasses the badge reader

○ B. Eavesdropping on wireless signals emitted by the servers

○ C. Denial-of-service attacks targeting the servers over the network

○ D. Social engineering of an employee in a common area

Correct answer: A

Why: Pressure sensors correlated with badge access logs detect presence in the room without a valid access event, meaning someone entered through a non-badged path (e.g., a ceiling, floor duct, or tailgating through an unmonitored route). This directly applies detection techniques to identify physical intrusion as described in LO 2.4.C.

*Why the others are wrong: **B**, Wireless eavesdropping is a logical/network attack vector; floor pressure sensors have no relevance to radio emissions. **C**, A DoS attack targets system availability through network channels; floor sensors cannot detect network-layer events. **D**, Social engineering occurs in human interactions, not within a physically secured server room.*

6. A security team wants evidence that can support a forensic investigation if a physical breach occurs. Which combination of controls best provides that evidence?

● A. Timestamped CCTV recording with 90-day retention and badge access logs that capture entry, exit, and failed attempts

○ B. Perimeter fencing and a roving security guard

○ C. Biometric entry scanner and mantrap vestibule

○ D. Cable locks on servers and privacy screen filters on monitors

Correct answer: A

Why: Forensic investigation requires a durable, timestamped record of events. Timestamped video recording with sufficient retention, combined with comprehensive access logs, provides two independent evidence streams. Biometrics and mantraps are preventive. Fencing and guards deter and delay but leave no persistent evidence record. Cable locks and screen filters protect against opportunistic theft but generate no logs.

*Why the others are wrong: **B**, A guard's observations are subjective and non-persistent; fencing has no recording capability. **C**, Biometric scanners and mantraps are strong preventive controls but do not by themselves create the timestamped evidence trail needed for forensic*

reconstruction. ***D****, Cable locks are physical theft deterrents; privacy screens protect visual confidentiality. Neither produces evidence of an intrusion event.*

7. During a physical vulnerability assessment of Xtensr Labs' newly acquired facility, you find: a badge reader that logs only successful entries, a CCTV camera covering the front entrance only, and no intrusion alarm system. Which gap in this detection setup creates the most significant risk to post-incident forensic analysis?

- **● A. The failure to log failed access attempts alongside successful ones**
- ○ B. The absence of perimeter fencing around the building
- ○ C. The lack of cable locks on server hardware
- ○ D. No documented visitor badge policy

Correct answer: A

Why: Failed access attempts are often precursor indicators of an attack, they show reconnaissance or trial-and-error. A system that logs only successes produces an incomplete audit trail: a successful breach is recorded, but the attacker's prior probing is invisible. NIST SP 800-53 PE-6 explicitly requires monitoring physical access to include attempts. Perimeter fencing is preventive, cable locks are asset-protection controls, and visitor policy is an administrative control, none directly impacts the forensic evidence record as critically as incomplete logging.

Why the others are wrong: ***B****, Fencing is a preventive perimeter control; its absence affects whether an attacker reaches the building, not whether an intrusion is recorded after the fact.* ***C****, Cable locks deter or slow hardware theft but produce no log or evidence chain if removed by force.* ***D****, A visitor policy is an important administrative control but its absence does not directly corrupt forensic evidence, the detection system still has its own gaps regardless.*

8. When evaluating the effectiveness of a physical monitoring system for a research lab, which metric most accurately measures how well the system identifies real intrusion events without generating excessive noise that causes security staff to ignore alerts?

- ○ A. Total number of sensors installed per square meter
- **● B. The system's detection rate measured against its false alarm rate**
- ○ C. The average response time of the on-call security officer
- ○ D. Per-unit cost of each installed sensor

Correct answer: B

Why: A high detection rate with a low false alarm rate indicates a system that reliably catches real events without generating so much noise that staff become desensitized and stop responding ("alert fatigue"). NIST SP 800-53 and physical security frameworks consistently evaluate detection effectiveness using the ratio of true positives to false positives, not sensor count, officer speed, or cost, all of which are secondary operational factors.

Why the others are wrong: ***A****, Sensor density affects coverage area but does not indicate whether the sensors are correctly calibrated to distinguish real events from background noise.* ***C****, Response time matters for incident response (a corrective/compensating function), not for measuring whether the detection system correctly identifies threats.* ***D****, Cost is a procurement criterion; it has no bearing on detection accuracy or reliability.*

9. A physical security analyst discovers that motion sensors in two adjacent server corridors have overlapping zones, yet an attacker who moves sufficiently slowly can pass through both zones without triggering either sensor, because each sensor's movement threshold is set for typical human walking speed. What class of vulnerability does this represent, and which enhancement best corrects it?

- ○ A. Sensor redundancy conflict; corrected by removing one sensor to eliminate zone overlap
- ○ B. Insufficient ambient lighting; corrected by installing brighter LED arrays in the corridors
- ○ C. Network misconfiguration in the sensor management console; corrected by applying the latest firmware patch
- **● D. Movement-speed threshold blind spot in motion detection; corrected by supplementing with passive infrared (PIR) heat-signature sensors and implementing cross-sensor event correlation**

Correct answer: D

Why: This is a sensor-capability vulnerability: microwave or ultrasonic motion sensors detect movement via reflected signal change, so a slow-moving intruder falls below the configured threshold. Supplementing with PIR sensors (which detect body heat independent of movement speed) and correlating events across both sensor types eliminates the speed-threshold blind spot. Removing overlapping sensors would reduce coverage. Firmware and lighting address separate, unrelated issues.

*Why the others are wrong: **A**, Sensor redundancy and overlapping zones are desirable, removing sensors would create gaps rather than fix the speed-threshold problem. **B**, Lighting improvements help CCTV image quality but do not resolve the motion detection threshold issue, a sensor blind to slow movement remains blind regardless of light levels. **C**, The vulnerability is in the physical sensor physics and threshold configuration, not in network or firmware settings.*

10. Which security strategy uses multiple overlapping detection layers so that if one layer fails to identify a threat, another layer remains in place to catch it?

○ A. Least privilege

○ B. Zero-day protection

● **C. Defense in depth**

○ D. Patch management

Correct answer: C

Why: Defense in depth (also called layered security) places multiple independent controls, perimeter, interior, and asset-level, so that compromising one does not expose the entire system. Least privilege limits user permissions. Zero-day protection addresses unknown software vulnerabilities. Patch management reduces known software vulnerabilities. None of these three describe the layered physical detection concept.

*Why the others are wrong: **A**, Least privilege is an access control principle governing permissions, not a physical detection architecture. **B**, Zero-day protection is specific to unknown software exploits; it has no physical layer analog in this context. **D**, Patch management is a vulnerability remediation practice for software systems, not a detection strategy architecture.*

11. A retail facility installs clearly visible CCTV cameras at every entrance and posts signs reading "This area is under 24-hour video surveillance." Beyond its detection function, what additional security role do these visible cameras serve?

● **A. Deterrence of potential attackers**

○ B. Data encryption of video feeds

○ C. Network segmentation of the camera system

○ D. Active prevention of physical intrusion

Correct answer: A

Why: Visible surveillance systems serve a dual role: they detect and record intrusion events, and they deter potential attackers by increasing the perceived risk of being identified. This is a well-established principle in physical security design, the camera's visibility is as strategically important as its recording function.

*Why the others are wrong: **B**, Encryption of video feeds is a data security practice for the camera system infrastructure, it is unrelated to the physical deterrence function of a visible camera. **C**, Network segmentation isolates systems on a network; it is a logical control, not a physical deterrence function. **D**, CCTV cameras record events but cannot physically block entry, prevention would require a door lock, barrier, or guard.*

12. Which of the following is NOT a physical detection control?

○ A. A passive infrared motion sensor

○ B. A door contact alarm that triggers when the door is opened

○ C. A CCTV camera with continuous recording

● **D. A reinforced steel deadbolt lock**

Correct answer: D

Why: A reinforced deadbolt lock is a preventive control, it resists or blocks unauthorized entry. Motion sensors, door contact alarms, and CCTV cameras are all detective controls: they identify that an event is occurring or has occurred but do not stop it from happening.

*Why the others are wrong: **A**, PIR motion sensors detect body heat and movement, a core physical detection technology. **B**, Door contact alarms trigger when a door is opened unexpectedly, generating an alert, clearly detective. **C**, CCTV with recording captures visual evidence of events, a detection and evidence-collection control.*

13. Xtensr Labs is designing camera placement for a server room that has a single badge-controlled entrance. The security goal is to detect both tailgating (a second person entering immediately behind an authorized user) and unauthorized removal of equipment from the racks. Which camera placement strategy best achieves both goals?

○ A. One camera mounted inside the room pointing at the server racks

● B. One camera facing inward at the entrance to capture all entrants, and one camera covering the server rack area

○ C. One camera mounted outside pointing at the entrance door

○ D. Cameras mounted only at exterior building perimeter entrances

Correct answer: B

Why: Tailgating occurs at the door transition point, requiring a camera that captures everyone who enters, regardless of badge status. Equipment removal occurs at the racks. Two cameras covering both locations provide overlapping but purpose-specific detection coverage. A single camera at either location leaves the other threat undetected. Exterior cameras only miss everything that happens inside.

*Why the others are wrong: **A**, An interior rack-facing camera would detect equipment removal but misses tailgating at the entrance entirely. **C**, An exterior entrance camera captures who approaches the door but cannot verify how many people enter during a single badge swipe. **D**, Exterior perimeter cameras support building-level detection but provide no visibility into server room activity.*

14. A physical security team is selecting a motion sensor for a storage area that houses active cooling equipment generating significant ambient heat. Which sensor type is most appropriate, and why?

○ A. Passive infrared (PIR) sensor, because it is the least expensive option

○ B. PIR sensor, because it provides a wider coverage angle than microwave sensors

● C. Microwave motion sensor, because it detects movement based on reflected signal change rather than heat signatures

○ D. Either type performs equally well; both sensors detect identical signals

Correct answer: C

Why: PIR sensors detect changes in infrared radiation, i.e., body heat. In an environment with high ambient heat from cooling equipment, the contrast between a human body and the surroundings is reduced, making PIR unreliable. Microwave sensors emit and detect reflected radio waves, detecting motion regardless of thermal environment. The choice is driven by the specific environmental conditions, not cost or angle alone.

*Why the others are wrong: **A**, Cost is a secondary consideration; selecting a cheaper sensor that generates frequent false positives (or misses detections) in a high-heat environment is a poor security decision. **B**, Coverage angle is a deployment consideration but does not address the core problem: PIR will perform unreliably in this thermal environment regardless of its angle. **D**, PIR and microwave sensors use fundamentally different physical principles and perform differently depending on environment, they are not interchangeable.*

15. An organization needs to detect if server hardware is physically removed from its rack without authorization. Which detection strategy most directly addresses this threat?

● A. Installing weight sensors in the equipment racks integrated with an asset management alert system

○ B. Encrypting the hard drives within each server

○ C. Requiring two-factor authentication for server room badge access

○ D. Scheduling weekly manual equipment inventory audits

Correct answer: A

Why: Weight sensors in racks detect the physical removal of hardware in near-real-time, and integration with an asset management system ties the alert to the specific asset that moved. This directly detects the physical attack event. Encryption protects data but does not detect that hardware was taken. 2FA controls who enters the room but not what happens inside it. Weekly audits would detect removal only up to a week after the fact, too slow for meaningful response.

*Why the others are wrong: **B**, Encryption is a data protection control; it renders stolen data less useful to an attacker but does not detect the removal event. **C**, Access control determines who may enter, but once inside legitimately (or after tailgating), no access control prevents or detects equipment removal. **D**, Manual weekly audits are better than nothing, but they are periodic detective controls, a theft could go undetected for days, making response effectively impossible.*

16. A physical security review of Xtensr Labs' newly acquired facility reveals: badge readers at the main entrance only, CCTV cameras covering public corridors but not the server or lab areas, and motion sensors that are disabled each evening to prevent false alarms triggered by the cleaning crew. Evaluate the most critical flaw in this detection strategy from a threat exposure standpoint.

● A. Disabling motion sensors after business hours creates a detection gap during the period when the facility is most vulnerable to undetected physical intrusion.

○ B. Badge readers at the main entrance should be upgraded to biometric scanners to reduce credential theft risk.

○ C. CCTV cameras in public corridors are redundant with badge readers and should be reallocated to server areas.

○ D. The facility should replace all electronic monitoring with dedicated on-site security personnel.

Correct answer: A

Why: Disabling motion sensors after hours eliminates detection capability precisely when the facility is least occupied and most attractive to intruders. High-value physical theft and sabotage events disproportionately occur outside business hours. The correct approach is to schedule the cleaning crew's access as a known exception within the monitoring system (or use a different sensor type less sensitive to cleaning activity) rather than disable detection entirely. The other options describe useful improvements but none introduces the same severity of exposure.

*Why the others are wrong: **B**, Upgrading to biometrics improves access control but does not address the complete absence of motion detection during off-hours, a more immediate gap. **C**, Public corridor and server area CCTV serve different detection purposes and are not functionally redundant. Removal of corridor cameras reduces coverage. **D**, Replacing all electronic monitoring with guards introduces human fatigue, coverage gaps, and higher operational cost, electronic systems complement rather than compete with personnel.*

17. NIST SP 800-53 control PE-6 requires organizations to monitor physical access to facilities. A security architect must design a detection strategy for a five-floor research building where the top two floors contain restricted research labs and the lower three floors contain general office space. Which approach best satisfies PE-6 while maintaining operational efficiency and proportionality?

○ A. Apply identical detection controls to all five floors regardless of sensitivity level to ensure consistent coverage.

● B. Apply tiered detection controls scaled to asset sensitivity: restricted lab floors receive CCTV, badge access logs, and motion sensors; general office floors receive CCTV coverage only.

○ C. Assign a dedicated security officer to each floor to eliminate the need for electronic monitoring.

○ D. Monitor only the building perimeter and rely on internal access controls to protect sensitive areas.

Correct answer: B

Why: PE-6 requires monitoring physical access but does not mandate identical controls everywhere, NIST frameworks are risk-based. A tiered approach applies more intensive detection where assets are more sensitive, satisfying PE-6 requirements for restricted areas while avoiding over-investment in lower-risk zones. Uniform controls applied everywhere regardless of risk wastes resources. Guard-only approaches are not scalable and introduce human error. Perimeter-only monitoring satisfies PE-6's boundary monitoring requirements but leaves interior high-value areas unmonitored.

*Why the others are wrong: **A**, Uniform controls satisfy PE-6 technically, but NIST frameworks explicitly encourage risk-based, proportional application, uniform application is inefficient and not best practice. **C**, Guard-only monitoring is not scalable across five floors 24/7, and human monitoring is subject to fatigue and error, making it an inferior primary strategy. **D**, Perimeter monitoring alone leaves the interior unmonitored. PE-6 requires monitoring of physical access to the facility, including internal sensitive areas.*

18. A workstation is stolen from Xtensr Labs over a weekend. A post-incident review finds: CCTV footage from the affected area is corrupted due to a storage failure, badge access logs were overwritten after 48 hours as configured, and motion sensor alerts were sent to a single email address belonging to a staff member who was on leave. Which systemic failure in the detection strategy does this incident most clearly expose, and what is the highest-priority remediation?

○ A. The CCTV cameras were insufficient resolution; replace with high-definition units to ensure footage quality.

○ B. Motion sensors should be replaced with pressure-plate sensors for greater accuracy.

○ C. The facility needed more security guards patrolling on weekends.

● D. The detection system lacked redundancy in alert routing and log retention; highest-priority remediation is implementing multi-channel alert escalation and extending access log retention to meet incident investigation timelines.

Correct answer: D

Why: Three failures converged: single-point-of-failure in alert delivery (one email address), inadequate log retention (48-hour overwrite that destroyed evidence within the investigation window), and no CCTV storage redundancy. These are systemic architectural failures, the detection system did identify the event, but the organization had no resilient way to act on or preserve that information. NIST SP 800-53 AU-9 (protection of audit information) and IR-6 (incident reporting) both require that evidence be protected and accessible. Camera resolution, guard staffing, and sensor type are secondary to fixing the fundamental monitoring architecture.

*Why the others are wrong: **A**, Resolution affects image clarity, but corrupted footage from a storage failure is a retention and redundancy problem, high-definition footage that is also corrupted provides no additional value. **B**, Sensor type selection addresses detection capability at the point of sensing; the failures here occurred downstream in alert delivery and evidence storage, not at the sensor itself. **C**, Additional guards address on-site human coverage but do not resolve the evidence preservation, alert routing, or log retention failures that prevented effective post-incident analysis.*